WOMEN INSTITUTE OF TECHNOLOGY AND INNOVATION

INTERNSHIP REPORT

CARRIED OUT AT

MINISTRY OF WORKS AND TRANSPORT IN NAKAWA, KAMPALA DISTRIT

(FROM 22ND JUNE, 2026 TO 22ND AUGUST, 2026)

BY

NAMONO FAITH

2024/DSC/0076/SS

A REPORT SUBMITTED TO THE WOMEN INSTITUTE OF TECHNOLOGY AND INNOVATION IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF A DIPLOMA IN COMPUTER SCIENCE

FIELD SUPERVISOR: MR. BUKENYA JONATHAN

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SUBMITTED: AUGUST 2026

DECLARATION

I, Namono Faith, do hereby declare that this Internship Report is a product of my own original work, carried out at the Ministry of Works and Transport, Nakawa Facility, during the months of June to August 2026 and that it has not been submitted before to any other institution of higher learning for the award of any academic qualification.

Signature: Date: 19th AUGUST 2026

Name: NAMONO FAITH

Registration Number: 2024/DSC/0076/SS

APPROVAL

This is to certify that this Internship Report by Namono Faith has been submitted for examination with the approval of the following supervisors:

Field Supervisor (Ministry of Works and Transport)

Name: _____

Signature: _____ Date: _____

Academic Supervisor (Women Institute of Technology and Innovation)

Name: _____

Signature: _____ Date: _____

ACKNOWLEDGEMENT

I would like to extend my heartfelt gratitude to the Almighty God Jehovah for the strength and guidance throughout my internship period.

My sincere appreciation goes to Women Institute of Technology and Innovation for the academic foundation and for organizing this internship programme, which enabled me to bridge the gap between classroom theory and practical, real-world application.

I am deeply grateful to the Ministry of Works and Transport for granting me the opportunity to undertake my internship in the IT support unit. Special thanks go to my field supervisor, Mr. Bukenya Jonathan, whose mentorship, patience and willingness to share knowledge greatly shaped my technical and professional growth during the internship period.

I am also immensely grateful to the members of the IT team- Ms.Byanyima Mellissa, Mr.Oketcho Denis, Ms.Nkata Gloria and Mr.Deogratious Ssempewo as well as other staff of the Ministry for their cooperation, guidance and the conducive working environment they provided throughout my attachment.

Finally, I extend my appreciation to my family and friends for their moral, financial and emotional support throughout my studies and internship period.

Table of Contents

DECLARATION 2
APPROVAL 3
ACKNOWLEDGEMENT 4
LIST OF TABLES 7
LIST OF FIGURES 8
LIST OF ABBREVIATIONS AND ACRONYMS 9
EXECUTIVE SUMMARY 10
CHAPTER ONE: INTRODUCTION 11
1.1 Background of the Internship 11
1.2 Objectives of the Internship 11
1.2.1 Specific Objectives 11
1.2.2 Training Objectives 11
1.3 Scope of the Internship 12
1.4 Methodology Used During Internship 12
CHAPTER TWO: PROFILE OF THE HOST ORGANIZATION 13
2.1 Introduction 13
2.2 Historical Background 13
2.3 Vision 13
2.4 Mission 13
2.5 Core Values 13
2.6 Organizational Structure 14
2.7 Main Functions and Services of the Organization 15
2.8 Departments/Units of the Organization 15
CHAPTER THREE: INTERNSHIP ACTIVITIES UNDERTAKEN 16
3.1 Introduction 16
3.2 Duties and Responsibilities Assigned 16
3.3 Activities Undertaken 16
3.3.1 Computer Hardware Maintenance and Diagnostics 17
3.3.2 Structured Cabling and Patch Panel Termination 21
3.3.3 Active Directory Domain Integration and Network Security 22
3.3.4 Tier 1 and Tier 2 Help Desk Support and Peripheral Maintenance 23
3.3.5 Biometric Attendance System Installation, Configuration and Enrolment 24
3.3.6 Printer Maintenance and Print Quota Management 26
3.3.7 Network Troubleshooting and IP Configuration 26
3.3.8 Testing Network Cables Using Cable Testers 27
3.3.9 Configuring Network Switches 28
3.3.10 Assigning IP Addresses 29
3.4 Skills and Knowledge Acquired 29
3.5 Relationship Between Academic Knowledge and Practical Experience 30
CHAPTER FOUR: CHALLENGES, LESSONS LEARNED AND RECOMMENDATIONS 31
4.1 Challenges Encountered 31
4.2 Lessons Learned 31
4.3 Recommendations 32
CHAPTER FIVE: CONCLUSION 33
5.1 Conclusion 33

REFERENCES 33
APPENDIX: PHOTOGRAPHIC IMPRESSION OF THE INTERNSHIP EXERCISE.....35

LIST OF TABLES

Table 1: Departments/Units of the Ministry of Works and Transport and their core functions

Table 2: Summary of IT support activities undertaken during the internship

Table 3: Challenges encountered and corresponding lessons learned

LIST OF FIGURES

Figure 1: Organizational structure of the Ministry of Works and Transport

Figure 2: Computer maintenance – front view of a serviced workstation

Figure 3: Computer maintenance – rear/internal view of a serviced workstation

Figure 4: Disassembled computer system unit

Figure 5: Physical inspection of hardware devices

Figure 6: Central Processing Unit (CPU)

Figure 7: Random Access Memory (RAM) module

Figure 8: Hard Disk Drive (HDD)

Figure 9: Ethernet cable termination (view 1)

Figure 10: Ethernet cable termination (view 2)

Figure 11: Assigning of IPv4 address parameters

Figure 12: Installation of Active Directory Domain Services (AD DS)

Figure 13: MPM (manufacturing programming mode) troubleshooting

Figure 14: Password reset using HB_CD software

Figure 15: Cabling work for biometrics installation

Figure 16: Configuring of biometrics device

Figure 17: Newly installed biometric terminal at the MoWT Kampala facility

Figure 18 & 19: Resolved printer errors

Figure 20: Crimping tool and cable tester

Figure 21: Motherboard

LIST OF ABBREVIATIONS AND ACRONYMS

Acronym	Full Definition
AD DS	Active Directory Domain Services
CAA	Civil Aviation Authority
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
HP	Hewlett-Packard
IT	Information and Communications Technology
IP	Internet Protocol
IT	Information Technology
WITI	Women Institute of Technology and Innovation
LAN	Local Area Network
WAN	Wide Area Network
MoWT	Ministry of Works and Transport
OS	Operating System
RAM	Random Access Memory
UEFI	Unified Extensible Firmware Interface
WAP	Wireless Access Point
VLSM	Variable Length Subnet Mask
UNRA	Uganda National Roads Authority

EXECUTIVE SUMMARY

This report presents my internship experience at the Ministry of Works and Transport, Nakawa Facility, during the months of June to August 2026.I am a diploma in science student at Women Institute of Technology and Innovation and I carried out internship under the supervision of Mr. Bukenya Jonathan in the IT Support Unit.

During my internship, I gained practical experience in IT support including hardware and software maintenance, network troubleshooting, printer maintenance, IT inventory management and preventive maintenance. I also gained experience in Active Directory, DNS, DHCP, access control, structured cabling, cybersecurity, antivirus management and IT equipment deployment.

The internship helped me apply the knowledge gained in class to a real working environment while improving my technical, communication and problem-solving skills.

CHAPTER ONE: INTRODUCTION

1.1 Background of the Internship

Internship is an essential component of the academic programme at Women Institute of Technology and Innovation, designed to provide students with an opportunity to apply theoretical knowledge acquired in class to real, practical work situations. It bridges the gap between academic learning and the demands of the professional job market, exposing students to the operational realities, work ethics, and technical demands of their respective fields of study.

As a requirement for the award of a Degree in Information Technology, I, Namono Faith, undertook a two-month internship at the Ministry of Works and Transport, Nakawa Facility, in the months of June to August 2026. This internship provided a valuable opportunity to gain hands-on experience in IT support services within a government institution.

1.2 Objectives of the Internship

General Objective

To gain practical experience and apply theoretical knowledge of Information Technology in a real work environment.

1.2.1 Specific Objectives

- To acquire practical skills in IT support, networking and systems administration related to the field of study.
- To understand the organizational operations, structure and culture of the Ministry of Works and Transport.
- To develop professional, technical, and interpersonal skills necessary for the IT job market.
- To identify challenges encountered in IT support work and propose practical solutions.
- To establish professional networks and working relationships within the IT industry.

1.2.2 Training Objectives

- To gain hands-on experience in computer repair, troubleshooting, maintenance and IT support.
- To gain hands-on experience in computer software installation, licensing, upgrade and use, driver installation and device activation.
- To understand the full lifecycle of IT infrastructure installation and support, particularly in network configuration and Internet.
- To enhance communication, teamwork and technical documentation skills within a professional IT setting.

1.3 Scope of the Internship

Geographical Scope: The primary base of the internship was carried out at the Ministry of Works and Transport, Nakawa Facility, Kampala, Uganda. In addition to the primary Nakawa site, the intern extended IT technical support and field maintenance services to other key ministry facilities including:

- **Kampala Facility (Industrial Area):** providing on-site hardware troubleshooting, hardware maintenance, and workstation domain connectivity.
- **Kireka Facility:** supporting network infrastructure, LAN maintenance and desktop system configurations for operational staff.
- **Mpigi Facility:** assisting in regional site IT support, enrolling MoWT staff on the employee attendance biometrics system.

Content Scope: The internship focused on IT support services, including help desk support, hardware and software maintenance, network administration, and systems support.

During the internship, the intern was exposed to a wide range of professional tasks related to the field of Computer Science. This opportunity was intended to provide hands-on experience that complements the theoretical knowledge gained during academic training at Women Institute of Technology and Innovation (WITI).

The intern's responsibilities spanned several functional areas, including installation and configuration of hardware and software components, operating systems, and key software applications, as well as support for network-related activities such as LAN setup, troubleshooting, and basic maintenance of connectivity infrastructure. This experience gave a practical understanding of how networks operate in a business environment.

The scope of the internship was not limited to technical work. The intern engaged in teamwork, received mentorship from senior IT officers and was introduced to workplace etiquette and professional communication. This well-rounded exposure greatly enhanced practical skills and broadened understanding of the real-world demands of the IT profession.

Time Scope: The internship covered a period of two months, from June to August 2026.

1.4 Methodology Used During Internship

In the course of the internship, several methods were used to acquire knowledge and skills, including:

- **Observation** — closely watching experienced IT technicians perform their duties before undertaking similar tasks independently, such as switch patching, server maintenance, and network routing configurations.
- **Participation** — actively taking part in day-to-day IT support activities such as assembling system units, directly handling helpdesk support tickets and connecting users to the MoWT domain under supervision.
- **Meetings and Consultations** — engaging in departmental briefings and consulting with the field supervisor and IT staff on technical matters.

CHAPTER TWO: PROFILE OF THE HOST ORGANIZATION

2.1 Introduction

This chapter presents a general overview of the Ministry of Works and Transport (MoWT), the host organization where the internship was undertaken, including its background, vision, mission, core values, organizational structure, and functions.

2.2 Historical Background

The Ministry of Works and Transport is a cabinet-level government ministry of the Republic of Uganda, headquartered along Old Port Bell Road in Kampala. The Ministry has for decades been responsible for the country's works and transport sector, overseeing the planning, development, and maintenance of road, rail, water, and air transport infrastructure, as well as public works such as government buildings. It provides policy and strategic oversight to several affiliated agencies, including the Uganda National Roads Authority (UNRA), the Civil Aviation Authority (CAA), the Uganda Road Fund, and the Uganda Railways Corporation (URC) (Ministry of Works and Transport, n.d.; Wikipedia contributors, 2025). The Nakawa Facility, where this internship was based, is one of the Ministry's operational stations supporting the IT and administrative functions of the institution.

2.3 Vision

To provide reliable and safe works, transport infrastructure and services for the people of Uganda (Ministry of Works and Transport, n.d.).

2.4 Mission

To promote the provision of adequate, safe, and well-maintained works and transport infrastructure and services for the social and economic development of Uganda (Ministry of Works and Transport).

2.5 Core Values

- Integrity and accountability in service delivery.
- Professionalism and technical excellence.
- Teamwork and collaboration.
- Innovation and continuous improvement.
- Client and public-focused service.

2.6 Organizational Structure

The Ministry of Works and Transport is headed politically by the Minister of Works and Transport, supported by State Ministers, with the Permanent Secretary providing technical and administrative leadership. Below this level, the Ministry is organized into directorates and departments such as Engineering and Works, Transport, Finance and Administration, and Planning, each further divided into units. The IT unit, under which this internship was undertaken, falls under the administrative support function and is responsible for maintaining the Ministry's information systems and technology infrastructure across its stations, including the Nakawa Facility.

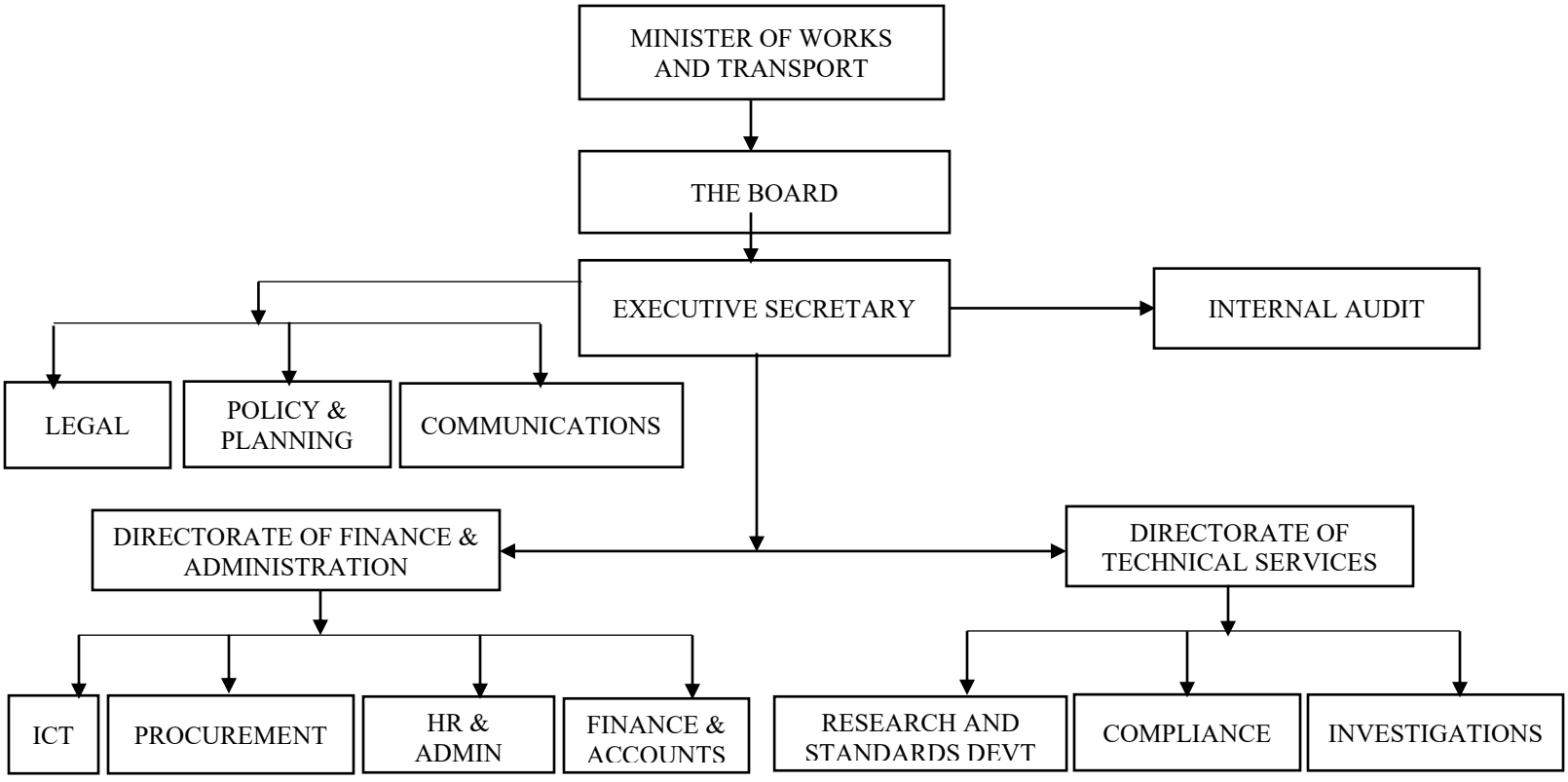


Figure 1: Organizational structure of the Ministry of Works and Transport.

2.7 Main Functions and Services of the Organization

As mandated under national government policy, the Ministry performs the following key functions (Ministry of Works and Transport, n.d.; Uganda National Web Portal, n.d.):

- Formulating policies related to works, transport, and engineering.
- Planning, developing, and maintaining road, rail, water, and air transport infrastructure.
- Managing public works, including government structures.
- Promoting quality standards in the construction industry.
- Providing regulatory and strategic oversight to affiliated transport and works agencies.

2.8 Departments/Units of the Organization

Table 1 summarizes the main departments and units of the Ministry, together with their core functions.

Department/Unit	Core Function
Directorate of Engineering and Works	Planning, design, and maintenance of public works and government structures.
Directorate of Transport	Policy formulation and coordination of road, rail, water, and air transport services.
Department of Finance and Administration	Financial management, budgeting, procurement, and general administration.
Department of Planning	Strategic planning, monitoring, and evaluation of Ministry programmes.
IT Unit	Maintenance of information systems, networks, and technology infrastructure.
Human Resource Department	Staff recruitment, welfare, training, and performance management.

Table 1: Departments/Units of the Ministry of Works and Transport and their core functions

CHAPTER THREE: INTERNSHIP ACTIVITIES UNDERTAKEN

3.1 Introduction

This chapter details the primary technical activities, system configurations, hardware operations, and network deployments executed during the field attachment at the Ministry of Works and Transport across the Nakawa, Industrial Area, Kireka, and Mpigi facilities. All procedures are presented using objective, system-focused technical descriptions.

3.2 Duties and Responsibilities Assigned

I was attached to the IT support unit of the Ministry of Works and Transport, Nakawa Facility, under the supervision of Mr. Bukenya Jonathan. My core responsibility was to provide IT support services to Ministry staff, ensuring that computer systems, network infrastructure, and peripheral devices remained fully functional to support the day-to-day operations of the institution.

3.3 Activities Undertaken

(Detailed day-to-day activities are recorded in the internship logbook.) Table 2 summarizes the key activities carried out during the internship period.

Activity	Description
Help desk support	Provided Tier 1 and Tier 2 support, responding to and resolving user-reported IT issues.
Hardware maintenance	Diagnosed and repaired faulty desktops, laptops, and peripheral devices.
Software installation	Installed and configured operating systems and application software.
IT inventory management	Maintained and updated the IT equipment register for asset tracking.
Network troubleshooting	Diagnosed and resolved LAN connectivity issues to restore user access.
Printer maintenance	Cleaned, serviced and resolved errors on office printers.
WAP installation	Installed and configured wireless access points to extend network coverage.
Print quota management	Managed print quotas to control departmental printing costs.
Preventive maintenance	Carried out routine servicing of computers and network equipment.

Activity	Description
AD DS administration	Assisted with user account creation and management in Active Directory.
DNS and DHCP support	Gained exposure to configuration and management of DNS and DHCP services.
Access control management	Supported management of access control for IT resources and facilities.
Structured cabling	Participated in cabling work and patch panel termination and organization.
Cybersecurity practices	Monitored for suspicious activity and safeguarded IT resources.
Antivirus management	Installed, updated, and managed antivirus software on Ministry computers.
Equipment deployment	Participated in deployment of new IT equipment to user departments.

Table 2: Summary of IT support activities undertaken during the internship

3.3.1 Computer Hardware Maintenance and Diagnostics

During the industrial training at the Ministry of Works and Transport, active participation was undertaken in computer hardware installation, maintenance, troubleshooting, and repair. These activities provided exposure to the internal architecture of desktop computers and facilitated the acquisition of practical skills in assembling, disassembling, upgrading, cleaning, and diagnosing hardware faults, restoring damaged units to operational status and optimizing system performance across regional facilities.

Prior to handling any computer equipment, thorough training was received on safety protocols, including disconnecting power sources, utilizing appropriate tools, and observing Electrostatic Discharge (ESD) precautions to protect sensitive electronic components.

Tools and Technologies Used:

- Precision screwdriver sets, anti-static wrist straps and compressed air blowers.
- Digital multimeters and power supply unit (PSU) testers.
- Replacement RAM (DDR3/DDR4), Solid State Drives (SATA SSDs) and thermal paste.

Step-by-Step Procedure:

- Workstation chassis were disconnected from power sources, opened, and internally cleaned using compressed air blowers to remove dust build-up.
- Defective internal components — including failing power supply units, degraded hard disk drives, and malfunctioning RAM sticks — were systematically isolated using diagnostic swaps.
- Thermal paste was reapplied on central processing unit (CPU) heatsinks to resolve thermal throttling issues.
- Storage capacity and system responsiveness were upgraded by installing new Solid-State Drives (SSDs) and inserting additional RAM modules into available DIMM slots.
- System units were reassembled, connected to peripherals, and subjected to power-on self-tests (POST) to verify hardware stability.

Outcome: Faulty workstation units at the Nakawa facility were restored to full functionality, yielding improved boot times and enhanced application processing speeds.

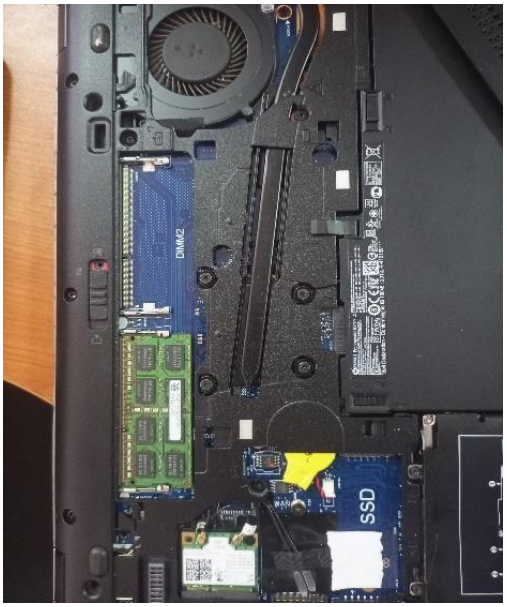


Figure 2: Computer maintenance – front view of a serviced workstation

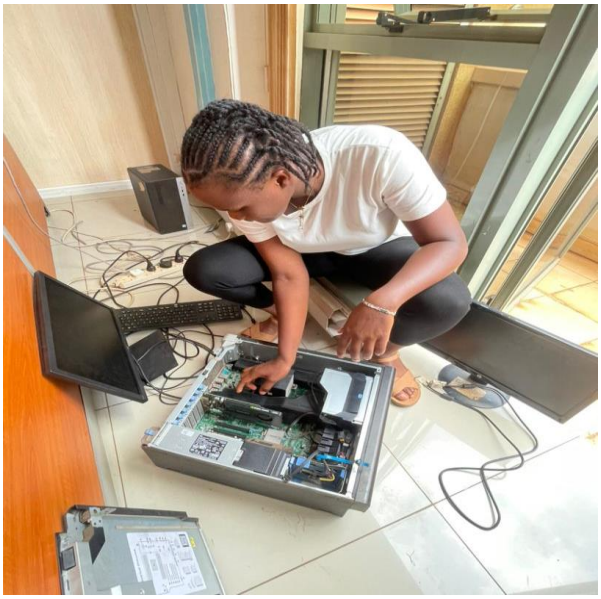


Figure 3: Computer maintenance – rear/internal view of a serviced workstation

Opening and Visual Inspection of Desktop Computers

One of the initial practical activities involved opening desktop computer system units. The process was initiated by disconnecting the computer from the electrical power supply and removing all external peripherals, including the monitor, keyboard, mouse, and network cables.

Using appropriate screwdrivers, the securing screws were removed, and the side panels of the computer casings were detached. Following the opening of each system unit, a visual inspection was conducted on the internal components to analyze their layout and identify visible faults, such as:

- Heavy dust accumulation
- Loose or disconnected cables
- Swollen or leaking capacitors
- Damaged connectors or signs of overheating

Proper procedures for handling internal components without causing physical damage or electrostatic discharge were strITly followed.

Tools Employed:

- Screwdrivers
- Anti-static wrist strap
- Cleaning brush
- Electric air blower
- Soft lint-free cleaning cloth

Skills Acquired:

- Safe procedures for opening and closing desktop system units.
- Proper handling techniques for sensitive electronic components.
- Identification and mapping of internal hardware layouts.



Figure 4: Disassembled computer system unit



Figure 5: Physical inspection of hardware devices

Identification of Motherboard Components

Following the opening of the computer systems, the functions and relationships of major motherboard components were analyzed. As the primary printed circuit board (PCB), the motherboard serves as the central hub interconnecting all hardware components. The key components identified and studied included:

Central Processing Unit (CPU): executed instructions and controlled processing tasks across the system.



Figure 6: Central Processing Unit (CPU)

Random Access Memory (RAM):

Serves as temporary, high-speed memory that stores data and instructions currently required by the CPU. RAM is volatile, meaning data is erased when power is disconnected. Higher RAM capacity enables smoother multitasking by reducing the CPU's reliance on slower secondary storage.

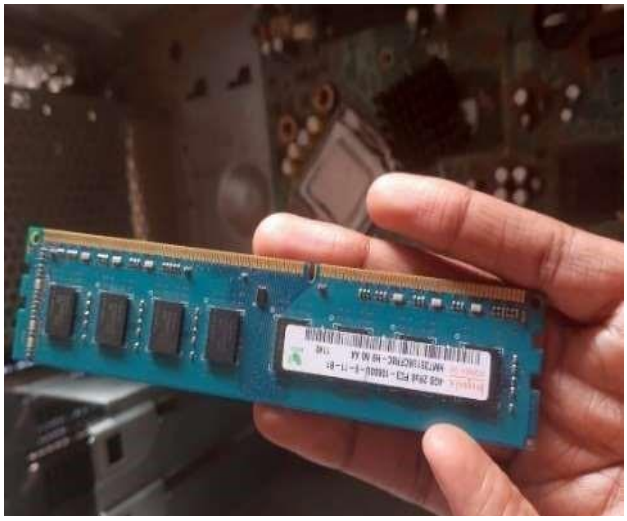


Figure 7: Random Access Memory (RAM) module

Hard Disk Drive (HDD):

Functioned as a secondary storage device that retains data magnetically on spinning platters accessed via read/write heads. It provides long-term, non-volatile storage where data remains saved even when the power is turned off.



Figure 8: Hard Disk Drive (HDD)

Skills Acquired:

- Systematic identification of core motherboard components.
- Comprehensive understanding of motherboard architecture and bus lines.
- Basic diagnostic techniques for motherboard-related faults.
- Practical knowledge regarding hardware compatibility standards.

3.3.2 Structured Cabling and Patch Panel Termination

Network patch cables were routed from server rack patch panels to managed network switches (switch patching) to establish active physical links, and Ethernet cables were terminated onto patch panels to create physical drop points in user offices.

Outcome: Physical network drops and patch panel links were successfully established, eliminating cable-related dead zones and securing high-speed local network connectivity across network racks.

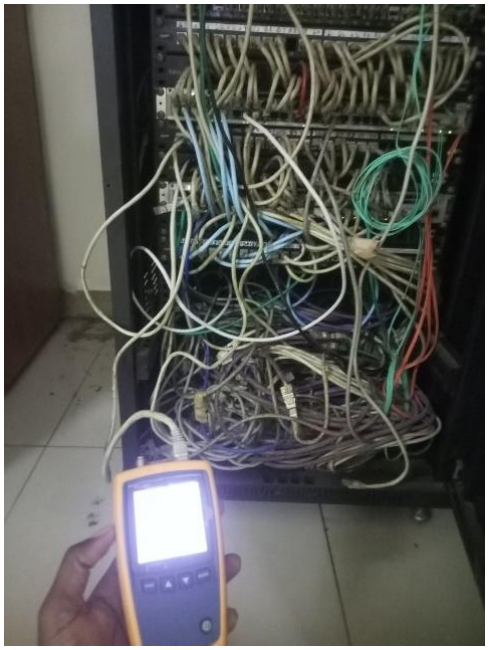


Figure 9: Ethernet cable termination (view 1)

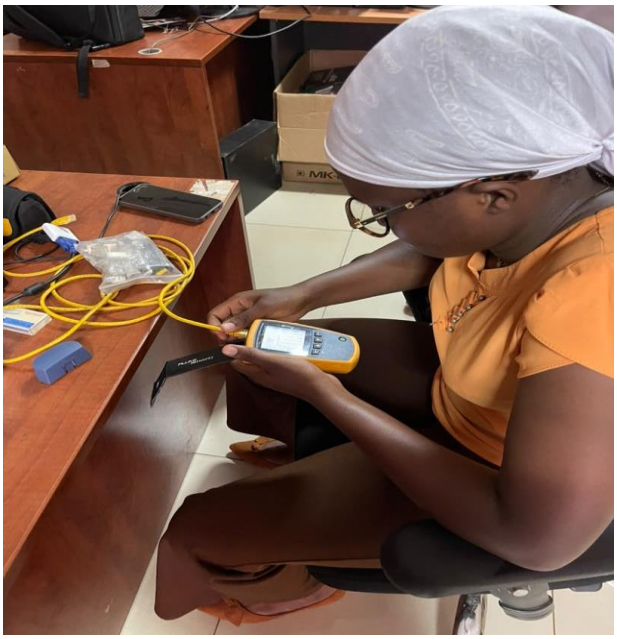


Figure 10: Ethernet cable termination (view 2)

3.3.3 Active Directory Domain Integration and Network Security

Task Overview: Workstations were integrated into the centralized Ministry of Works and Transport (MoWT) Active Directory domain to enable centralized access control, user authentication, and network resource sharing.

Tools and Technologies Used:

- Windows Enterprise / Professional edition operating systems.
- MoWT Active Directory Domain Controller infrastructure.
- Command Prompt (CMD) network utilities (ipconfig, ping, nslookup).

Step-by-Step Procedure:

- Network Adapter properties were accessed to configure TCP/IPv4 parameters, assigning static IP addresses, subnet masks, default gateways, and local DNS server addresses.
- Network connection status and domain name resolution were validated via Command Prompt using ping and nslookup commands.
- System Properties were navigated on target workstations to modify workgroup settings, selecting the option to join an existing domain.
- The fully qualified domain name was entered, followed by authentication using domain administrator credentials.
- Workstations were restarted, domain user accounts were assigned, and local administrative permissions were applied as governed by organizational access policies.

Outcome: User workstations across different facilities were successfully joined to the MoWT domain, establishing authenticated domain login access, printer sharing, and centralized security policy compliance.

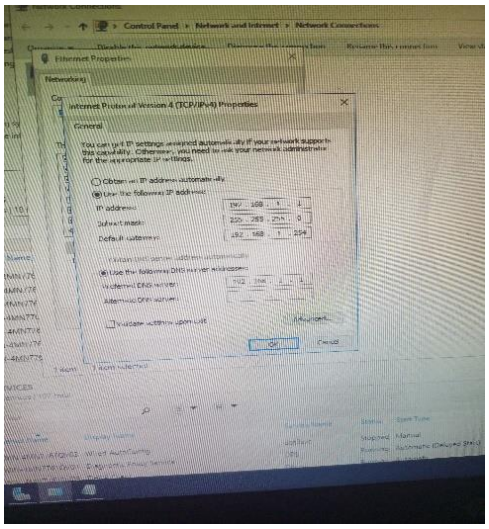


Figure 11: Assigning of IPv4 address parameters

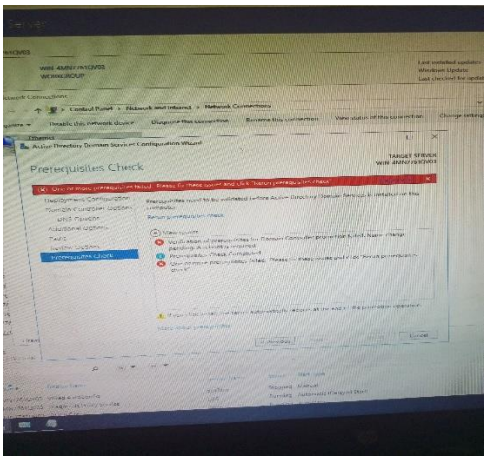


Figure 12: Installation of Active Directory Domain Services (AD DS)

3.3.4 Tier 1 and Tier 2 Help Desk Support and Peripheral Maintenance

Task Overview: First-line technical helpdesk operations were managed to provide user support, resolve software application errors, fix print server connectivity, and clear operational bottlenecks for staff.

Tools and Technologies Used:

- Helpdesk ticketing records and remote desktop support utilities.
- Network-shared laser printers and driver utilities.
- Operating system repair command line utilities (sfc /scannow, chkdsk).

Step-by-Step Procedure:

- Support calls and helpdesk requests were logged, categorized by severity, and prioritized based on operational impact.
- Software application errors and corrupted system files were evaluated and repaired using system file checker (sfc /scannow) routines.
- Network printer connectivity was configured by mapping shared network IP addresses, installing appropriate print drivers, and running test pages.
- User password reset requests and domain account lockouts were resolved in coordination with senior IT officers.
- Technical ticket resolutions were recorded, verified with end-users, and officially closed in the support log.
- Forgotten passwords were reset by me using password reset software.
- Computer errors such as MPM (manufacturing programming mode) were resolved to prevent system downtime.

Outcome: Technical support requests at Nakawa were systematically resolved, minimizing downtime for operational staff.

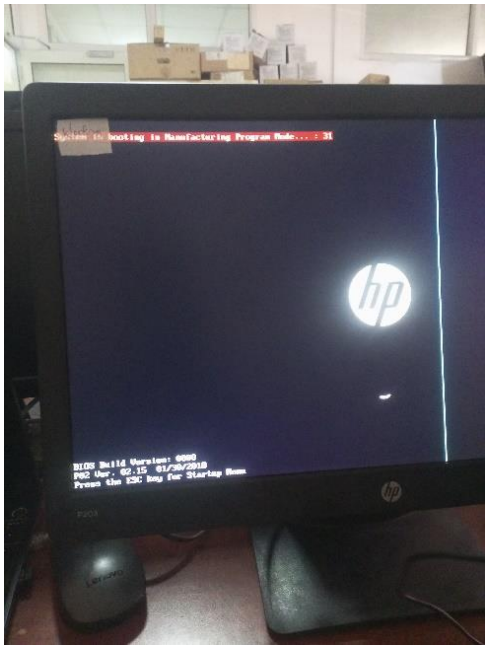


Figure 13: MPM (manufacturing programming mode) troubleshooting

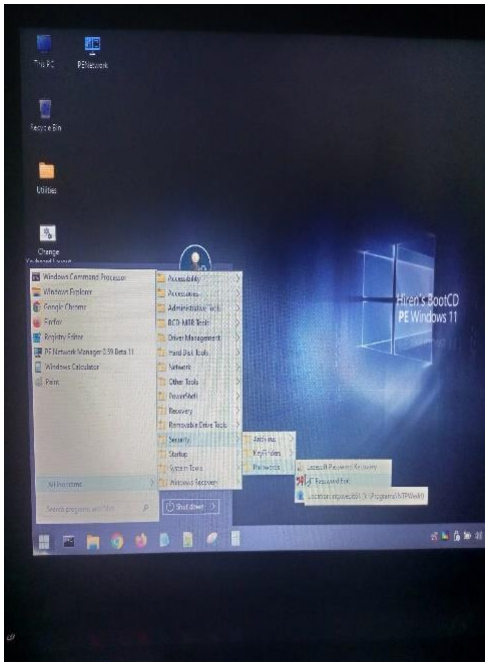


Figure 14: Password reset using HB_CD software

3.3.5 Biometric Attendance System Installation, Configuration and Enrolment

Task Overview: Biometric access control and time-attendance terminal systems were installed, configured, and integrated into the Ministry's local area network to digitalize attendance tracking, secure facility access, and enable user biometric registration across ministry facilities.

Tools and Technologies Used:

- Standalone biometric fingerprint/face recognition terminals.
- Ethernet patch cables (CAT6) and RJ45 connectors.
- Biometric management software.
- TCP/IPv4 configuration utilities and mounting software.

Step-by-Step Procedure:

- **Physical and network installation:** biometric hardware terminals were wall-mounted at designated site entrances, connected to local power supplies and linked to the facility's local area network using CAT6 UTP cables.
- **Network parameter configuration:** static IPv4 address parameters (IP address, subnet mask, default gateway) were manually assigned directly via the terminal menu interface to ensure static network connectivity.

- **Software setup and device communication:** the biometric management software was installed on the central system server and communication parameters (IP address, port number, device ID) were configured to establish connection between the server and remote terminals.
- **User biometric enrolment:** Ministry staff members were registered into the system by entering employee ID details, scanning multiple primary finger templates via the optical scanner, and capturing face signatures.
- **Data synchronization and verification:** test transactions were performed on the terminal to verify scan accuracy, followed by running automated data synchronization routines to transfer event logs from terminals to the central management database.

Outcome: Biometric terminals were successfully installed and networked, with staff members at the MoWT headquarters enrolled in the database, establishing automated and accurate attendance and access control management.



Figure 15: Newly installed biometric terminal at the MoWT Kampala facility



Figure 16: Configuring of biometrics device

3.3.6 Printer Maintenance and Print Quota Management

Objective/Task Overview: Printer maintenance and print quota management were carried out to keep office printers in good working condition and to control printing costs across departments.

Tools and Technologies Used: Printer driver software, replacement toner/ink cartridges, and print management software.

Step-by-Step Procedure:

- Printers were inspected for paper jams, worn rollers, and low toner or ink levels.
- Print heads and rollers were cleaned, and consumables were replaced as required.
- Printer drivers were reinstalled or updated where printing errors persisted.
- Print quotas were configured and monitored using print management software to regulate the number of pages printed per department.

Outcome: Printer downtime was reduced, and departmental printing costs were brought under better control.

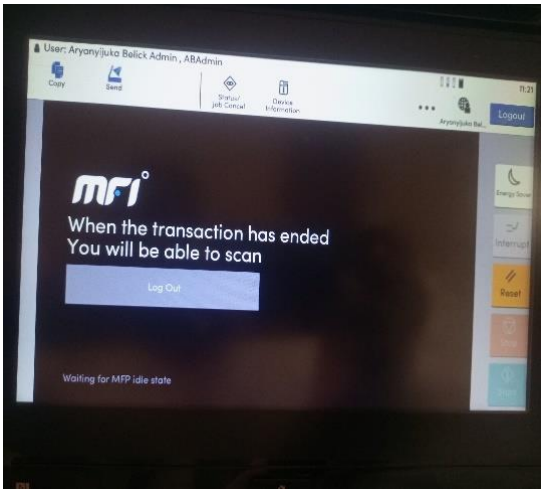




Figure 18 & 19: Resolved printer errors

3.3.7 Network Troubleshooting and IP Configuration

Objective/Task Overview: Network troubleshooting was carried out to diagnose and resolve local area network (LAN) connectivity problems affecting staff workstations.

Tools and Technologies Used: Network cable tester, ping and tracert command-line utilities, and the Control Panel network settings.

Step-by-Step Procedure:

- Reported connectivity issues were investigated by checking physical cable connections and network adapter status.
- The Control Panel was accessed, and the Network and Sharing Center was opened to review adapter settings.

- Internet Protocol Version 4 (TCP/IPv4) properties were opened, and the static IP address, subnet mask, and default gateway details provided by the network administrator were input where required.
- Connectivity was tested using ping commands via the command prompt to confirm successful communication with the network gateway and other hosts.

Outcome: Network connectivity was restored, and workstations were able to communicate reliably within the LAN and access shared resources.

3.3.8 Testing Network Cables Using Cable Testers

After crimping Ethernet cables, I tested them using a network cable tester. The tester was used to verify continuity between the two ends of the cable and ensure that each wire was correctly connected to the corresponding pin.

Whenever faults such as crossed pairs, open circuits, or short circuits were detected, I identified the problem, re-crimped the connectors where necessary, and repeated the test until the cable passed successfully.

This activity enhanced troubleshooting skills and emphasized the importance of testing before deploying network cables.



Figure 20: Crimping tool and cable tester

3.3.9 Configuring Network Switches

Network switches were configured to facilitate communication among computers within a Local Area Network (LAN). This involved connecting network cables to the appropriate switch ports, accessing the switch configuration interface, assigning management IP addresses, and ensuring proper port connectivity.

I learned how switches forward data based on MAC addresses, reducing unnecessary network traffic and improving communication efficiency.

Through this activity, I developed practical skills in installing and configuring switches used in modern computer networks.

3.3.10 Assigning IP Addresses

Another activity I carried out assigning IP addresses to computers and other network devices. This involved configuring both static and dynamic (DHCP) IP addresses depending on the network requirements.

I configured important network parameters including the IP address, subnet mask (which identifies which part of an IP address is the network and which part is the host), default gateway, and preferred DNS server. After assigning the addresses, I verified connectivity using commands such as ipconfig, ping and tracert. This activity enabled a clearer understanding of IP addressing, subnetting concepts and communication within computer networks.

3.4 Skills and Knowledge Acquired

Technical Skills

I acquired hands-on technical skills in computer hardware and software maintenance, network troubleshooting, structured cabling, Active Directory, DNS and DHCP administration, wireless network installation, and cybersecurity practices, including antivirus management.



Figure 21: Motherboard

Communication Skills

Handling helps desk support requests improved the ability to listen to user problems, explain technical issues in simple terms, and communicate solutions clearly and professionally.

Teamwork Skills

Working closely with the IT team on shared tasks, such as equipment deployment and structured cabling, strengthened the ability to collaborate effectively with colleagues to achieve common goals.

Problem-Solving Skills

Diagnosing and resolving hardware, software, and network faults sharpened analytical and troubleshooting abilities under real work conditions.

Leadership and Management Skills

Managing the IT inventory and coordinating print quota allocations gave practical exposure to resource management and accountability.

Time Management Skills

Balancing multiple help desk tickets and maintenance tasks within tight timelines improved the ability to prioritize and manage time efficiently.

3.5 Relationship Between Academic Knowledge and Practical Experience

The internship provided a practical platform to apply concepts learned in class, such as computer networks, systems administration, and database and directory services, in a real institutional setting. For instance, classroom knowledge of networking protocols was directly applied while troubleshooting LAN connectivity issues and configuring DHCP and DNS, while theoretical understanding of directory services was reinforced through hands-on work with Active Directory Domain Services. This experience confirmed the relevance of the Computer Science curriculum and highlighted areas where further practical exposure would be beneficial.

CHAPTER FOUR: CHALLENGES, LESSONS LEARNED AND RECOMMENDATIONS

4.1 Challenges Encountered

- Limited spare parts and replacement components, which occasionally delayed the repair and servicing of faulty computer hardware.
- Inadequate IT equipment and tools to handle the high volume of help desk requests promptly.
- Time constraints in resolving multiple concurrent support tickets, especially during peak periods of network downtime.
- Difficulty adapting to the formal government work environment and its administrative procedures during the early weeks of the internship.
- Financial and transport challenges in commuting to and from the Nakawa Facility on a daily basis.
- Occasional resistance from some staff members when troubleshooting required temporary disruption of their work.

4.2 Lessons Learned

Table 3 links the key challenges encountered during the internship to the corresponding lessons learned.

Challenge Encountered	Lesson Learned
Limited spare parts and equipment	The importance of resourcefulness and proper planning in IT support work.
High volume of concurrent support tickets	Practical problem-solving and prioritization under pressure.
Adapting to a formal government environment	Professional ethics, conduct, and confidentiality in handling data.
Coordinating tasks with the IT team	The value of teamwork in resolving complex technical issues.
Communicating with non-technical staff	Improved communication skills in explaining technical issues simply.
Managing IT inventory and access	Responsibility and accountability over shared organizational resources.

Table 3: Challenges encountered and corresponding lessons learned

4.3 Recommendations

To the Host Organization

- Provide closer mentorship to interns at the start of the internship period.
- Continue to maintain and potentially increase the frequency of structured feedback sessions, allowing interns to understand their progress and areas for improvement more regularly, building on the positive feedback observed in the logbook.
- Continue to provide access to IT tools and equipment and internal documentation and if possible, curate a list of external online learning platforms or courses relevant to the tools and technologies used to support intern self-learning.

To the Institute

- Increase collaboration and partnerships with host organizations such as the Ministry of Works and Transport.
- Integrate more hands-on practical workshops into the curriculum, particularly those that simulate real-world IT project scenarios and utilize industry-standard tools, hardware, software, and network configuration equipment. This would better prepare students for the technologies they encounter in internships.
- Ensure the internship activity logbook format is clear, consistent, and provides ample space for detailed weekly reporting of activities, lessons, challenge and skills as this proved invaluable for report generation.
- Explore possibilities of providing some form of support or guidance for students facing logistical challenges such as transport and daily meals during their internship period.

To Future Interns

- Be proactive, punctual and maintain professionalism throughout the internship period.
- Observe organizational policies and seek guidance from supervisors whenever necessary.
- Be Adaptable: The IT field is constantly evolving. Be open to learning new technologies and adapting to different work environments and project requirements. Embrace challenges as learning opportunities.
- Embrace Self-Learning: The internship is a prime opportunity for self-directed learning. Utilize online resources, documentation and the expertise of colleagues to expand your knowledge and skills rapidly especially when encountering new technologies.
- Network Effectively: Build relationships with colleagues and supervisors and importantly with other interns. These connections can be invaluable for future career opportunities, mentorship and gaining deeper industry insights especially when direct supervision is limited.
- Document Everything Diligently: your weekly activity logbook is your most valuable asset for the internship report and future career reflections. Fill it out meticulously, detailing tasks, challenges, lessons and skills gained.

CHAPTER FIVE: CONCLUSION

5.1 Conclusion

The internship undertaken at the Ministry of Works and Transport, Nakawa Facility, from June to August 2026, was a rewarding and enriching experience that provided practical exposure to real-world IT support operations within a government institution. Working under the supervision of Mr. Bukenya Jonathan, I was involved in a wide range of activities, including Tier 1 and Tier 2 help desk support, hardware and software maintenance, network troubleshooting, wireless access point installation, Active Directory, DNS and DHCP administration, structured cabling, cybersecurity practices, and IT equipment deployment.

Through this internship, I acquired valuable technical, communication, teamwork, problem-solving, leadership and time management skills that will be instrumental in a future career as an Information Technology professional. Despite the challenges encountered, such as limited resources and time constraints, the overall experience strengthened the link between academic knowledge and practical application, and significantly contributed to academic and professional development.

In conclusion, it is recommended that such internship programmes continue to be supported by both Women Institute of Technology and Innovation and host organizations, as they play a vital role in preparing students for the demands of the professional IT job market.

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APPENDIX: PHOTOGRAPHIC IMPRESSION OF THE INTERNSHIP EXERCISE




NAKAWA
BUSINESS PARK
BLOCK C
DIRECTORATE OF ROAD INFRASTRUCTURE
PROTECTION
IT DEPARTMENT